



# Benchmarking Depth-Sensor 3D Phenotyping of Maize Cobs Against 2D Image and Geometric Methods

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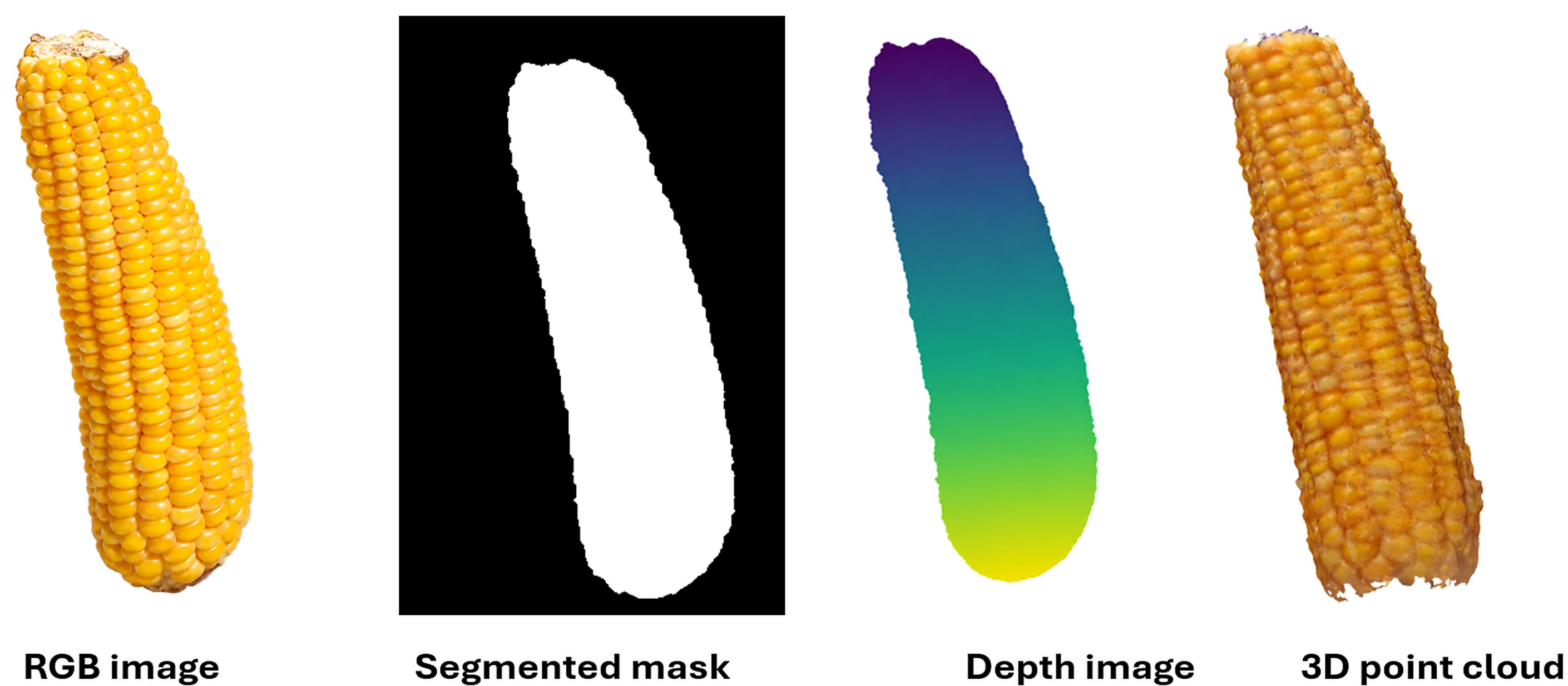
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## Introduction

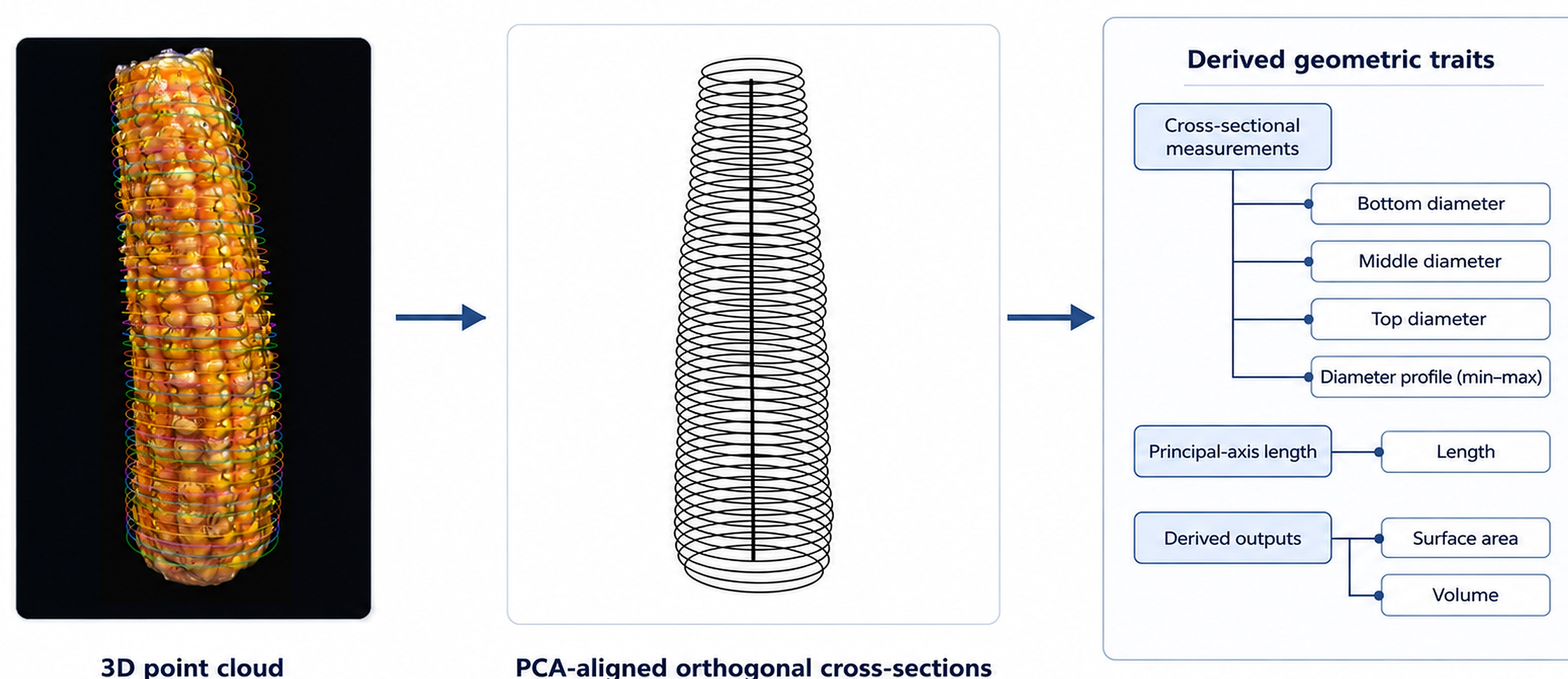
Conventional maize cob phenotyping commonly relies on manual caliper measurements with simplified cylindrical assumptions or on 2D RGB image analysis. These approaches can introduce bias when cobs are curved, tapered, or irregular in shape. This study benchmarks a low-cost depth-sensing 3D phenotyping pipeline against 2D image-based and geometric methods for estimating maize cob morphological traits. The main imaging outputs used in the workflow are shown in **Figure 1**.



**Figure 1.** Depth-sensing outputs for maize cob phenotyping

## Method

- RGB-depth data were used to reconstruct a 3D cob point cloud (**Figure 1**).
- PCA alignment and orthogonal slicing generated ring-wise diameter profiles (**Figure 2**).
- 3D traits were benchmarked against 2D projection and geometric methods (**Table 1**).



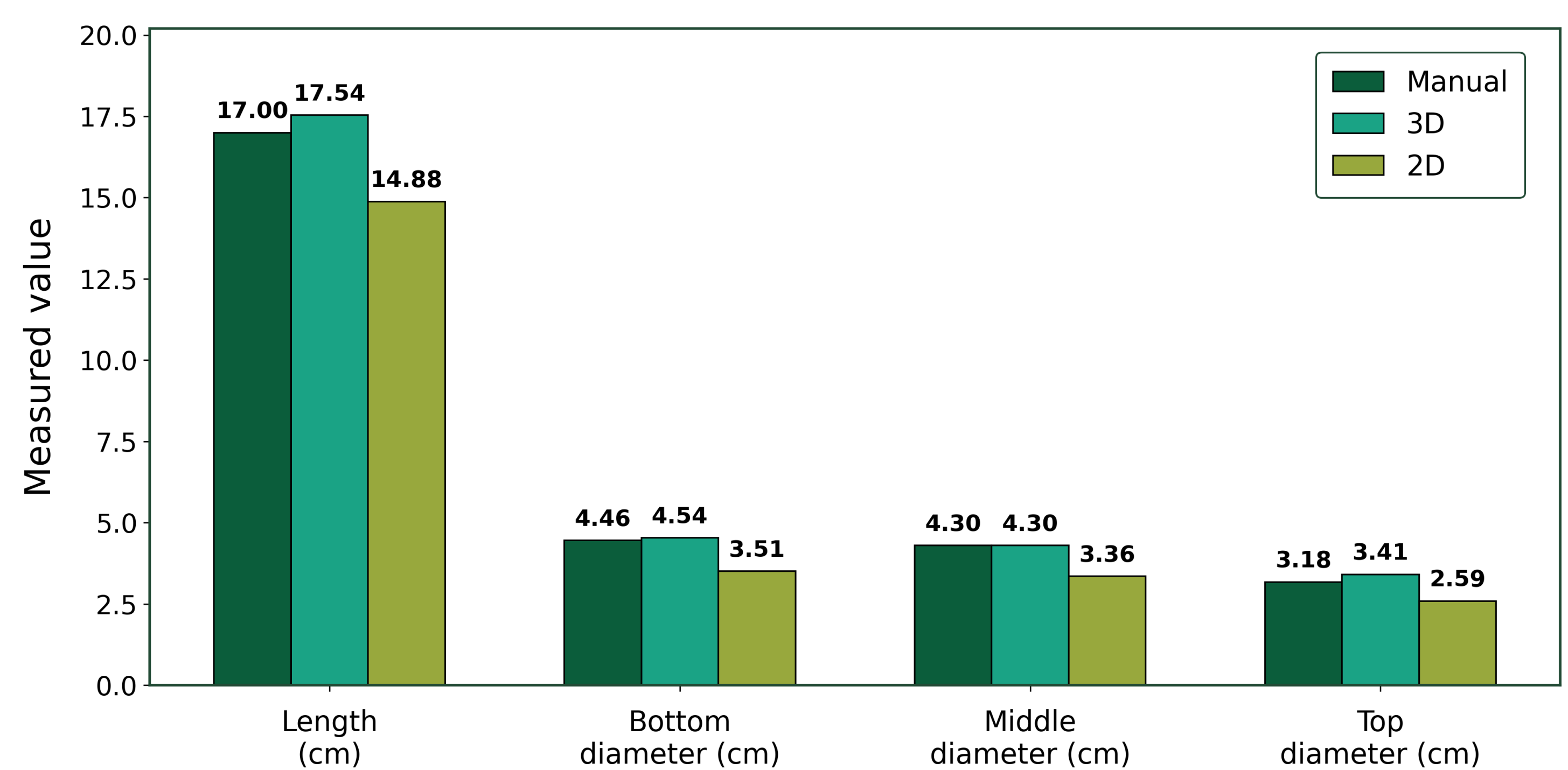
**Figure 2.** Workflow for extracting maize cob traits from a PCA-aligned 3D point cloud

## Results

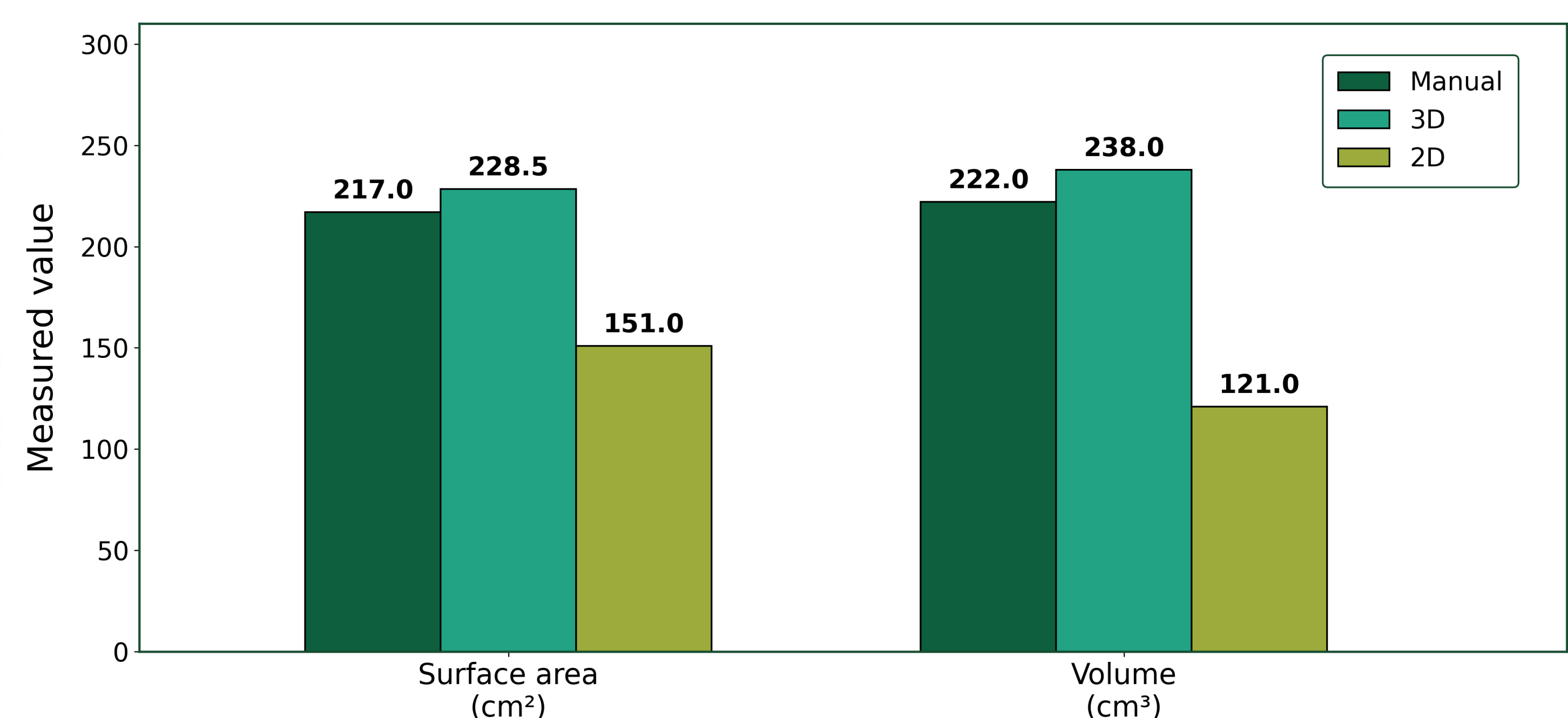
- The 3D method showed closer agreement with reference measurements (**Table 1**).
- It reduced errors for length, diameter, surface area, and volume (**Figures 3–4**).
- 3D profiling better captured tapered and irregular cob shapes.

**Table 1.** Comparison of manual reference measurements with 3D depth-sensor and 2D image-based estimates

| Trait                           | Manual measurement | 3D estimation | 2D estimation |
|---------------------------------|--------------------|---------------|---------------|
| Length (cm)                     | 17.00              | 17.54         | 14.88         |
| Error                           |                    | 3.2%          | 12.5%         |
| Avg diameter (cm)               | 3.98               | 4.08          | 3.15          |
| Error                           |                    | 2.6%          | 20.8%         |
| Volume (cm <sup>3</sup> )       | 222.0              | 238.0         | 121.0         |
| Error                           |                    | 7.2%          | 45.5%         |
| Surface area (cm <sup>2</sup> ) | 217.0              | 228.5         | 151.0         |
| Error                           |                    | 5.3%          | 30.4%         |
| Mean error                      |                    | 4.6%          | 27.3%         |



**Figure 3.** Manual, 3D, and 2D estimates for cob length and local diameters



**Figure 4.** Manual, 3D, and 2D estimates for cob surface area and volume

## Conclusion

- Low-cost depth sensing improves maize cob phenotyping.
- 3D diameter profiling reduces bias from 2D projection and cylindrical assumptions.
- The method supports scalable cob trait estimation.